# Carrier board v1: soldering guide

Board 154 x 96 mm, bare copper (no printing). Every part goes on the TOP; every pin is soldered on the BOTTOM. Pages 2 and 3 show both sides, page 4 what to do if the holes are not copper-plated.

## Which side is which

1. Lay the board so the row of 4 large 5-pin footprints (module plugs J1-J4) is along the edge nearest you.
2. TOP (parts side): the long double row of 24 pads (Teensy socket) is at the LEFT, starting at the left edge; the short double row of 6 (DAC socket) is right of it, near the far edge. The 2 left mounting holes are lower (32 and 58 mm from the far edge) than the 2 right ones (10 and 45 mm).
3. If the Teensy rows are at the right instead, you are looking at the BOTTOM (solder side).
4. Check against page 2: the tracks you see on the top must match the red tracks there.

## Before you start

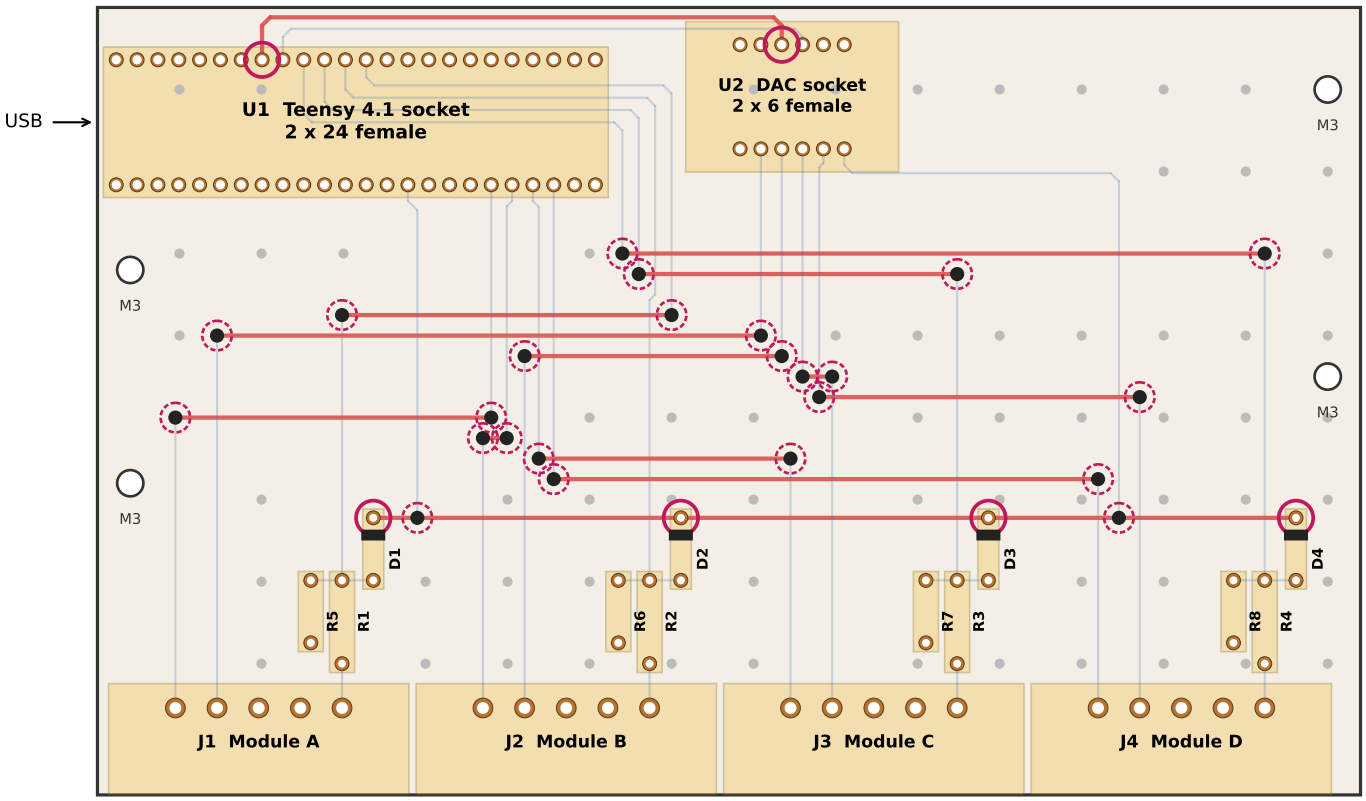
1. Look into a few holes. Copper-lined (shiny ring inside) = plated: solder on the bottom only. Bare fibreglass inside = NOT plated: also do page 4.
2. The board takes FEMALE headers (sockets). The Teensy and the DAC keep their male pins and plug in from the top. Never solder male pins into the board, and never solder the Teensy or DAC to it.
3. Multimeter in beep mode at hand. No Teensy, no DAC, no modules on the board while soldering.

## Parts and order (lowest first, so each part lies flat on the bench while you solder)

| # | Where | Part | How |
|---|---|---|---|
| 1 | R1-R4 | 4 x 1 kOhm (brown-black-red) | either way round; the long 10 mm ones |
| 2 | R5-R8 | 4 x 1 MOhm (brown-black-green) | either way round; the short 7.6 mm ones |
| 3 | D1-D4 | 4 x BAT85 diode (glass) | BAND towards the far edge (Teensy side): page 2 |
| 4 | U1 | 2 x female header 1x24, 8.5 mm | Teensy socket, two rows |
| 5 | U2 | 2 x female header 1x6 | DAC socket, two rows |
| 6 | J1-J4 | 4 x Phoenix MSTB 2,5/5-GF-5,08 | plug opening towards the board edge |

# 2. TOP side: where every part goes

Seen from above, parts side up, module plugs at the bottom of the picture. Push each part in from this side; its pins come out on the bottom.



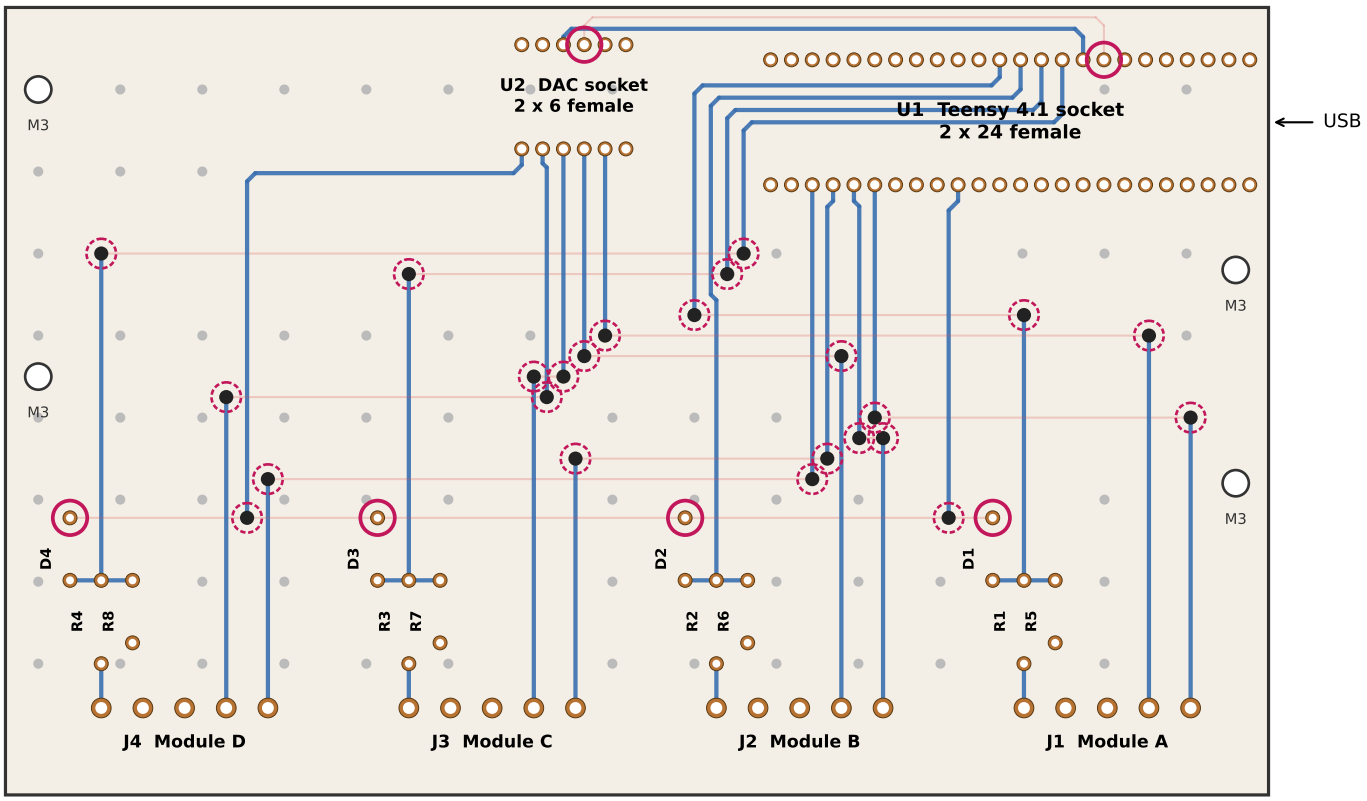

module plugs: this edge (cables plug in here)

where the part body sits
top copper track (on this side)
bottom track (other side, seen through)
pad with hole (one pin each)
via: track changes side here (small grey dots: ground vias, nothing to do)
circled: needs solder ON TOP too, if the holes are not plated (page 4)

1. Resistors and diodes: bend the leads to the hole spacing, push them in until the body lies on the board, bend the leads out a little underneath so they stay, solder, cut.
2. Diodes D1-D4: the black band (black bar in the picture) goes to the hole nearer the far edge (the 3.3 V line). Wrong way round, it shorts the force signal to 3.3 V.
3. Sockets: plug the female headers onto the Teensy (and the DAC) or onto spare male pin strips, set them in, solder one pin at each end, check they sit flat and straight, then solder the rest.
4. MSTB plugs J1-J4: the opening where the cable plug goes in faces the near board edge; the flange screws are at its ends.

# 3. BOTTOM side: where you solder

The board turned over (left and right swap: the Teensy is now on the RIGHT). Every pad here gets one solder joint: heat pad and pin together 2-3 s, feed solder until it forms a small cone, cut the leads ~1 mm.




module plugs: this edge (cables plug in here)

- bottom copper track (on this side)
- top track (other side, seen through)
- pad with hole (one pin each)
- via: track changes side here (small grey dots: ground vias, nothing to do)
- circled: needs solder ON TOP too, if the holes are not plated (page 4)

1. Small joints, no bridges: socket pins are only 2.54 mm apart. After each row, look at it against light.
2. Ground pads sit in the copper fill and take longer to heat: hold the iron 1-2 s more.
3. Circled pads and vias: if the holes are not plated, they also need the top-side work on page 4.

# 4. Only if the holes are NOT plated

In a plated board the copper inside each hole joins the top and bottom. Without it, a joint on the bottom does not reach a top track. Two places need extra work (circled in pink on pages 2 and 3):

## A. 6 pads with a track only on the TOP: solder them on the top as well

- D2 pad 1  (diode, band end): +3V3
- U1 pad 41  (Teensy socket, Teensy pin 19): SCL
- D1 pad 1  (diode, band end): +3V3
- U2 pad 3  (DAC socket, its SCL pin): SCL
- D4 pad 1  (diode, band end): +3V3
- D3 pad 1  (diode, band end): +3V3

Diodes: easy, the leads are reachable; solder that lead on the top too. Sockets U1 / U2: the plastic covers the top pad. Set the header in, but leave it raised ~1 mm; solder that pin on the top first, then push the header down and solder all pins underneath. Or solder a thin wire from the top pad to the pin.

## B. 24 signal vias: a short wire through each, soldered on both sides

A via is where a track goes from one side to the other. Push a cut-off resistor lead through it, solder on top and bottom, cut both ends short. The grey vias (ground) can be left: ground is joined by the copper fill on both sides.

- +3V3: (125, 62) mm,  (39, 62) mm
- EN_A: (48, 50) mm,  (10, 50) mm
- EN_B: (50, 52) mm,  (47, 52) mm
- EN_C: (54, 55) mm,  (84, 55) mm
- EN_D: (122, 58) mm,  (56, 58) mm
- F_A: (30, 38) mm,  (70, 38) mm
- F_C: (66, 32) mm,  (105, 32) mm
- F_D: (142, 30) mm,  (64, 30) mm
- SP_A: (15, 40) mm,  (81, 40) mm
- SP_B: (52, 42) mm,  (83, 42) mm
- SP_C: (90, 45) mm,  (86, 45) mm
- SP_D: (127, 48) mm,  (88, 48) mm

Positions in mm from the top-left corner seen from the top (USB edge left, plug edge at the bottom). Check each with the multimeter afterwards: beep from the pad on one side to the track it should reach on the other.

# 5. Check before plugging anything in

Multimeter on beep. No Teensy, no DAC, no modules (README 'Assembly and first power-up').

1. 3V3 to GND must NOT beep: Teensy socket 3V3 pin to any J pin 3 or 4.
2. Each plug J1-J4: pins 3 and 4 to a Teensy socket GND pin must beep.
3. J1 pin 1 to Teensy socket pin 27 must beep; J2 pin 1 to 28, J3 to 29, J4 to 30 (module enable).
4. DAC socket SCL to Teensy pin 19 and SDA to pin 18 must beep (these use the top track and a via: page 4).
5. Each J pin 2 (setpoint) to its DAC output pin must beep (A-D: through vias).
6. Then plug in the DAC and the Teensy (USB at the board edge), USB only: bumpem info --port <PORT> must answer.
7. One module at a time, 48 V off: plug in, then scripts/open_loop_test.sh for that module.